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```
class Device {
    String type = "Basic";

    public void method1() {
        method2();
        System.out.println("Device m1");
    }
    public void method2() {
        System.out.println("Device m2");
    }

    public String toString() {
        method2();
        if(this instanceof Device){
            return "Device is " + type;
        }
        else{
            return "Device is not " +
type;
        }
    }
}
class Laptop extends Device {
    String type = "Portable";

    public void method2() {
        System.out.println(this.type);
        System.out.println("Laptop m2");
    }

    public void method3() {
        System.out.println(((Device)this).
basic);
        System.out.println("Laptop m3");
    }
}
class Phone extends Device {
    String type = "Smart";

    public void method1() {
        System.out.println("Phone m1");
    }

    public void method4() {
        System.out.println("Phone m4");
    }

    public String toString() {
        method2();
        return "Phone is " + type;
    }
}
```

```
class MacBook extends Laptop {
    String type = "Apple";

    public void method1() {
        super.method1();
        System.out.println("MacBook m1");
    }

    public void method3() {
        System.out.println("MacBook m3");
        System.out.println((ProMac(this)).type);
    }
}
class ProMac extends MacBook {
    String type = "Pro";

    public void method1() {
        System.out.println("ProMac m1");
    }

    public void method4() {
        System.out.println("ProMac m4");
    }

    public String toString() {
        method2();

        System.out.println(((Laptop)this).method4());
        return "ProMac is " + type;
    }
}
```

Assuming the following variables have been defined:

```
Device dev1 = new Device();
Phone phone1 = new Phone();
Laptop lap1 = new Laptop();
Laptop lap2 = new MacBook();
Object obj1 = new Laptop();
Object obj2 = new Phone();
Device pro1 = new ProMac();
```

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	Statement	Output
1	dev1.method1();	
2	lap1.method1();	
3	lap2.method4();	
4	pro1.method1();	
5	System.out.println(pro1);	
6	System.out.println(pro1.type);	
7	((Device)phone1).method2();	
8	((MacBook)lap2).method3();	
9	((Laptop)phone1).method2();	
10	((Device)phone1).method3();	
11	((Device)lap2).method1();	
12	((Device)obj1).method2();	
13	((ProMac)obj1).method2();	
14	((MacBook)obj2).method3();	
15	System.out.println(((Object)dev1).toString());	
16	System.out.println(((Device)phone1).type);	